

High resolution spatio-temporal modelling of solar photovoltaic potential for tropical islands: Case of Mauritius

Jay Rovisham Singh Doorga^{a,*}, Soonil D.D.V. Rughooputh^a, Ravindra Boojhawon^b

^a Department of Physics, Faculty of Science, University of Mauritius, Reduit, Mauritius

^b Department of Mathematics, Faculty of Science, University of Mauritius, Reduit, Mauritius

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ABSTRACT

Tropical and subtropical islands benefit from a wide spectrum of renewable energy resources and advancement in technology has made it feasible for harnessing these clean and abundant energy sources. Mauritius being located in the tropical belt, receives plentiful sunshine all year round. Consequently, to provide assistance to the Government and other relevant organizations wishing to deploy photovoltaic facilities on the island, the methodological framework and results presented have been organized to serve as a strong background to identify the optimum sites for efficient harnessing of solar energy through photovoltaic systems. The Multiple Criteria Decision Making (MCDM) analysis coupled to Analytical Hierarchy Process (AHP) have been used to model the solar photovoltaic potential of the island from a set of social, technical, legal, environmental and climatological factors. The performance of the model has been calibrated using the monthly fluctuations in electricity production from a photovoltaic power plant located on the island. We propose an adapted mathematical formula to theoretically estimate the electricity production from photovoltaic systems on the island on the temporal basis. The outcome of this research is the production of a high resolution yearly mean solar photovoltaic map alongside with the spatio-temporal variability of photovoltaic potential throughout the year.

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1. Introduction

The geographical location of the Republic of Mauritius within the tropical belt ensures abundance of renewable energy resources. The island benefits throughout the year from a diverse spectrum of clean energy resources that range from the profusion of solar energy, persistent south-east trade winds and high occurrence of wind-induced waves off its coastal morphology. Despite the tremendous wealth of its energetic natural resources, the country is still heavily dependent on fossil fuel import, mostly from the Middle East, for electricity production. A share of 2379 GWh, contributing worth 78% of the electricity generation in 2016, was derived mostly from coal and fuel oil [1]. Consequently, the economic fluctuations on the international markets of fossil fuel prices, bear significant repercussions on the financial structure of the country. The Government recognized the potential of renewable energy resources and is actively engaged in the promotion of

sustainable measures. This led to the introduction of attractive policies and the allocation of a capital budget for renewable energy development.

Among the measures taken to bolden the sustainable energy transition, lies the formulation of schemes allowing the progressive market penetration of photovoltaic (PV) infrastructure in the form of Small Scale Distributed Generator (SSDG) feed in tariff mechanism and Medium Scale Distributed Generator (MSDG) net metering system. Another manoeuvre includes the setting up a network of solar farms on the island, with the recent issue of tenders for a total of 62 MW for solar farms to be commissioned in 2018–2019 [2]. Additionally, another project aims at equipping the houses of about 10,000 low income families with photovoltaic panels [3]. In the policy document entitled 'Long Term Energy Strategy 2009–2025', the Government of Mauritius shares its vision to increase renewable energy integration to 35% by 2025 [4]. Renewable energy has the capacity to meet the increasing energy demands while keeping carbon emissions to a safe level. The current challenge however is to provide reliable affordable energy while transitioning towards renewable energy sources. Solar energy is revealed to be the right candidate for a Small Island

* Corresponding author.

E-mail addresses: jay.doorga927@gmail.com (J.R. Singh Doorga), rughooputh.sddv@gmail.com (S.D.D.V. Rughooputh), r.boojhawon@uom.ac.mu (R. Boojhawon).

Developing State (SIDS) like Mauritius so as to refrain from relying on remote countries in order to satisfy its local energy needs. This is highly encouraged by the favourable solar climate during the two seasons of the year.

Mauritius is a developing country with significant pressure on limited land resources. The population of Mauritius stood at 1,263,820 in 2016, representing a growth of 0.1% as compared to the previous year [5]. The increasing population dynamics coupled with limited land resources are vital factors that need to be taken into consideration so as to ensure sustainability. In SIDS, pressures on land resources exacerbated through their competitive usage, growing demands and degrading land among others [6]. Consequently, it is of paramount importance to provide appropriate legislative and regulatory frameworks for the coordinated management of the island's land resources. International strategies to provide assistance to SIDS and disseminate best practices and technologies for implementing sustainable land-use policies have been conveyed in the 2009 report of the United Nations Conference on Environment and Development [6]. As such, identification of the optimum solar energy harnessing locations would help provide strategic plans and incentives for efficiently harnessing energy from the sun while delimiting solar panels arrangement to a localized area, thereby minimizing land occupation and construction cost.

High penetration of solar energy on international markets, achieved through the management of its intermittent nature, has been made possible with the constantly-improving technologies and the adoption of appropriate operational strategies. The year 2016 has seen record numbers regarding newly installed renewable power generating capacity, with solar photovoltaic (PV) representing about 47% of newly installed renewable power capacity [7]. To achieve energy autonomy, it is imperative for islands situated near tropical and subtropical latitudes to embrace solar energy technologies. Tropical and subtropical regions witness low seasonality within their solar radiation which enable high share renewable energy resources coupled with short term energy storage technologies [8]. The Canary Islands, located in the Atlantic subtropical belt, installed worth 112 MW of solar PV back in 2010 with the objective to attain 30% electricity from renewable energy resources [9]. Sister Island, Reunion, produced 60 GWh electricity from stand-alone and interconnected PV systems in the same year [10]. In 2014, the Hawaiian islands of Oahu achieved 10% solar energy penetration from tropical solar microclimates [11].

This paper focusses on the spatial distributions of solar energy resources in Mauritius, in order to detect the spatially more promising regions on the island to concentrate ground-mounted monocrystalline silicon PV systems. The objective of this study is to locate propitious areas on the island for photovoltaic systems integration and construction. With this purpose, a high resolution solar PV resource potential map of Mauritius (1:300,000 scale) is presented. The variations of the parameter on the temporal scale is also explored with the objective to device proper maintenance strategies on potential solar farms and other PV systems. The justification of the study conducted is to be able to efficiently harness electricity from the sun while delimiting solar panels to a confined location. This study is expected to save on land resources of an island state through the identification of hot spots for photovoltaic placements.

The solar energy resource map is implemented in the Geographical Information System (GIS) platform and a Multi-Criteria Decision Making (MCDM) analysis is performed in order to reveal high potential zones for exploitation. MCDM problems involve a set of alternatives that are evaluated on the basis of conflicting and incommensurate criteria [12]. Factors which have observed to influence the electricity generation from PV systems

are considered. The weight attributed to each factor, highlighting its relative importance, is determined through the Analytical Hierarchy Process (AHP). Regions deemed unfavorable for development are excluded from the analysis surface through the implementation of constraint factors in the model.

The novelty of this study is that it employs a mathematical formula that takes into account the effect of temperature on solar photovoltaic panels' efficiency and generates a calibrated model describing the variations of solar photovoltaic potential on micro-scale levels. The importance of this study is that it offers island states, with land scarcity problems, a way to identify the optimum regions for harnessing solar energy efficiently through the delimitation of solar panels arrangements to a confined area. Consequently, the methodology described in the upcoming sections will help island states and other countries around the globe to identify regions that are forbidden for the development of photovoltaic systems in addition to identifying the best and less favourable regions.

For the proposed methodology, this work presents an important step in modelling the spatio-temporal distributions of solar PV potential in Mauritius. The study may be useful to the Government and other relevant organizations in making an informed decision pertaining to the placement of PV systems in the country.

The methodological framework of this article is as follows: Section 2 presents the study area, flowchart methodology adopted, Geographical Information System (GIS) used, constraint factors, evaluation criteria and theoretical background. The latter incorporate the AHP, Weighted Linear Combination (WLC) and Inverse Distance Weighting (IDW) processes. The results of the spatial variations in the solar photovoltaic potential of Mauritius and corresponding temporal fluctuations as a function of the month of the year is presented in section 3. Also present in this section is the access to high voltage transmission lines. Finally, the conclusive remarks of the study are presented in section 4.

2. Methodology

2.1. Study area

The South West Indian Ocean (SWIO) Island State of Mauritius is situated off the eastern coasts of Africa (Fig. 1(b)) and consists of a major island, situated at 20.3° South and 57.6° East together with a group of outer islands and islets including Rodrigues, Agalega, Cargados Carajos shoals, Chagos Archipelago and Tromelin. Of volcanic origin, the main island of Mauritius (Fig. 1(a)) was formed during the early Pliocene about 7.8 million years ago and gave rise to a relief comprising of a high upland central plateau and low lying areas at near sea level [13]. Roughly elliptical in shape, Mauritius measures about 60 km long and 45 km wide. The island incorporates in its morphological structure, a group of isolated peaks, mountain chains and variable land slopes.

Mauritius possesses a maritime climate that varies from tropical during summer (November to April) to sub-tropical during winter (May to October). Average temperatures are 20.4 °C in winter and 24.7 °C in summer, though many coastal regions experience higher temperatures than these thresholds (sometimes reaching 36 °C in January and February) [14]. The suitable microclimate systems prevailing at spatially distinct sites on the island favour the implementation of PV systems. The solar climate of the island is found to be influenced by spatially clustered cloud cover zones generated by the variable topography and giving rise to a yearly average daily global solar irradiation of about 16 MJ/m²day [15]. The settlement patterns in Mauritius is portrayed in Fig. 1(c), with the densest population concentrations situated in the districts of Plaines Wilhems and Port-Louis.

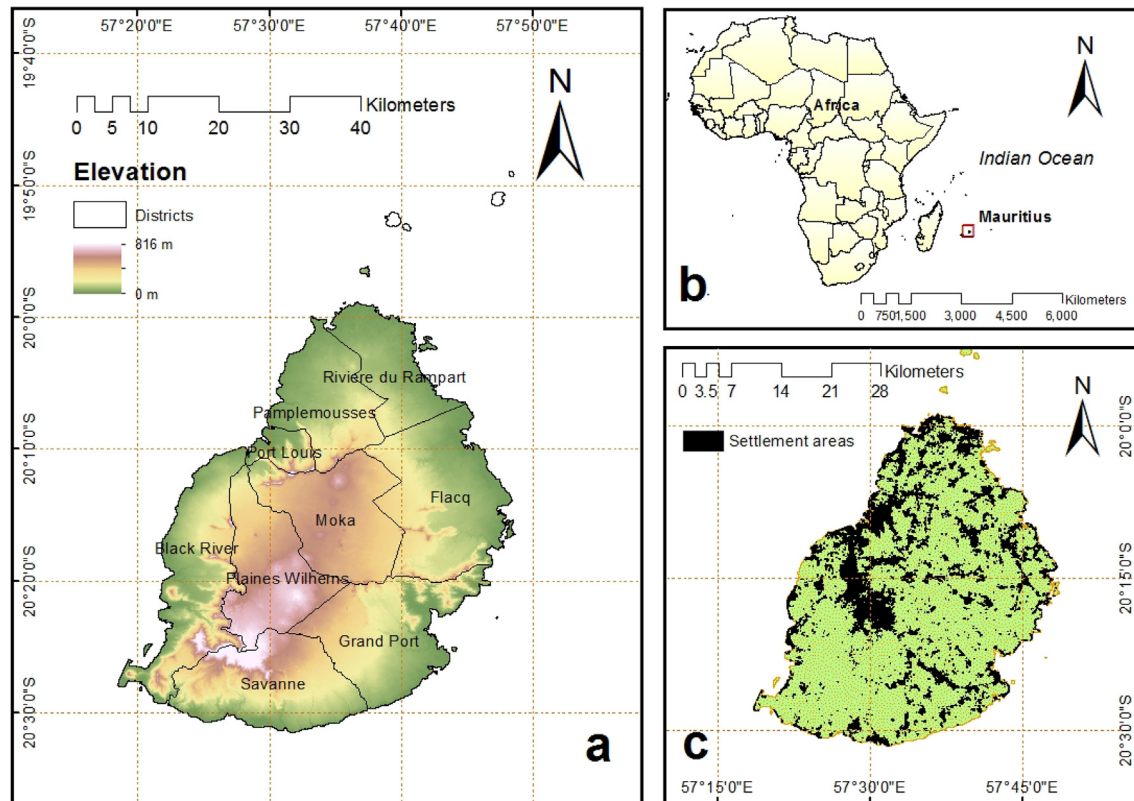


Fig. 1. (a) Digital Elevation Model (DEM) of the island of Mauritius with demarcation lines representing the different districts. (b) Geographical location of Mauritius in the south-west Indian Ocean basin. (c) Map of Mauritius delineating the settlement areas on the island.

2.2. Material and methods

2.2.1. Flowchart methodology

Locating the most appropriate regions for ground-mounted PV facilities is dependent on a large number of functional parameters. The methodological framework provided in Fig. 2 delineates the steps undertaken in a strategic manner as presented in this paper. Factors considered as constraints to the placement of PV systems are removed from the study area. A digital geo-database, derived from the DEM and comprising of shadow zones and exclusionary slope regions, are used. Additionally, spatial information on World heritage sites, nature reserves, botanical gardens and wildlife protected sites are employed.

The next stage involves setting up a digital database of evaluation criteria consisting of conditional and influential factors. PV installation sites are preferably constructed near settlement areas in addition to favourable installations in proximity to existing road and grid systems. Consequently, we cater for these identified conditional factors in the present study. Influential factors on the other hand are the most important criteria in determining the best regions for harnessing solar energy through PV systems. As influential factors, the sunshine duration as well as the electricity generated from PV facilities are explored.

After having obtained the list of criteria influencing the placement of PV systems on the island, the relative weights attributed to each one of them is determined through the Analytical Hierarchy Process (AHP). Classification is then conducted, prior to performing the weighted overlay process in the GIS platform. The most suitable regions for potential exploitation will be thereafter revealed.

2.2.2. The Geographical Information System (GIS)

The Geographical Information System (GIS) employed to

model the photovoltaic potential of Mauritius is illustrated in Fig. 3. The GIS incorporates both data collection and organization phases. The data used in the current study were either made available, derived from existing data or generated using GIS specialized software. Digital Elevation Model (DEM), settlement area, meteorological data, road and grid networks were made available by the relevant authorities of the country as mentioned in the upcoming texts. Slope, aspect and shadow zone variables were derived from the high resolution DEM of the island. Finally, ArcMap 10.3 was used to digitize spatial constraints including World Heritage sites, nature reserves and wildlife zones. Digitization of geographical features was performed using the process of georeferencing and creation of shapefiles under the ArcToolbox environment using the ArcGIS software as summarized by Doorga et al. (2018a) [16].

The data were arranged into two main groups which include the constraint and evaluation sets as shown on the right hand side of Fig. 3. The constraint set contains all data regarded as a restriction to the placement of photovoltaic systems. The evaluation set on the other hand contains all data that influences the placement of photovoltaic systems on the island. After the evaluation set data are used to build up the model, the constraint set data are overlain on the GIS solar photovoltaic map to give the suitable and unsuitable regions for photovoltaic project construction.

2.2.3. Contrast among models used in literature

A contrast among GIS models employed in literature to map the photovoltaic resource potential at country and regional scales is presented in Table 1. GIS models employed in literature are diverse, yet quite distinctive and make use of several input parameters. A summary is provided here alongside the proposed model.

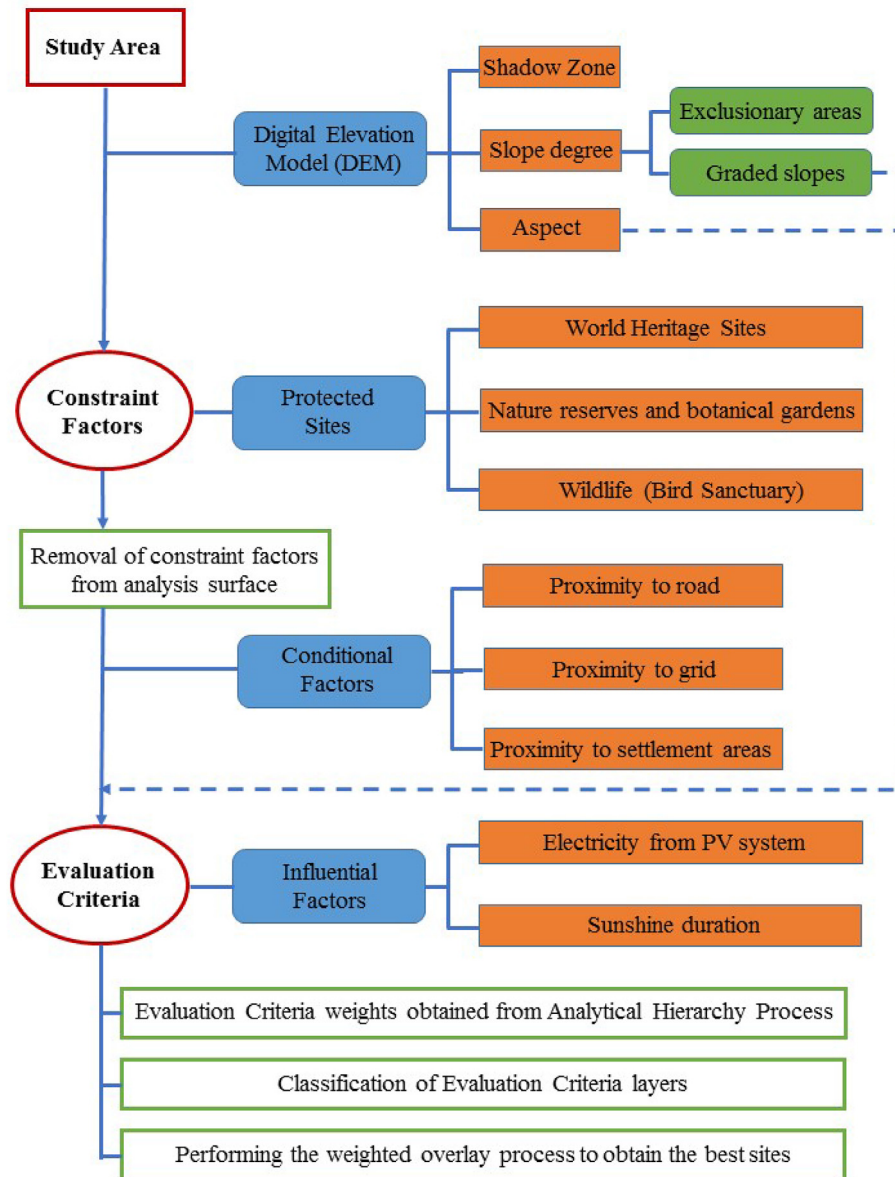


Fig. 2. Flowchart with methods demonstrating the various stages involved in determining the PV potential of Mauritius.

2.2.4. Constraint factors

Criteria regarded as constraints for the placement of ground-mounted PV infrastructure are discussed in this section.

2.2.4.1. World Heritage Sites. Mauritius is home to two UNESCO World heritage sites: Le Morne Cultural Landscape and Aapravasi Ghat (Fig. 5). Since its inscription as a UNESCO World Heritage Site in 2008, Le Morne Brabant is widely considered as a symbol of marronage and freedom during the nineteenth and twentieth centuries in Mauritius [23]. Runaway slaves fled to seek refuge in the caves of Le Morne Mountain while many of them threw themselves off its summit to their deaths to prevent being captured. Aapravasi Ghat on the other hand was proclaimed as World Heritage Site in 2006 and is regarded as an emblem of indentured servitude, which marks the immigration portal where the immigrants disembarked [24]. The majority of them arrived at the immigration depot where they were processed by British authorities prior to being sent on sugarcane plantations. Due to the deep

historical nature of both sites, legal frameworks have been implemented to protect these zones through the implementation of the National Heritage Fund Act 2003 [25]. The buffer zones of both sites have been acquired from the Ministry of Housing and Lands.

2.2.4.2. Biosphere conservation. The wildlife of the island, comprised of its flora and fauna, is under close supervision by local conservation authorities. Human activities in the country have led to the destruction of many natural habitats and threatened to the point of extinction many endemic species. Consequently, in an attempt to protect the remaining parcels of the original forests on the island, strict laws have been set up to prohibit development in the buffer zones. Protection involves seven nature reserves (Perrier, Les Mares, Gouly Pere, Cabinet, Bois Sec, Le Pouce and Corps de Garde); two national parks (Black River Gorges and Bras d'Eau); two botanical gardens (Pamplemousses and Curepipe Botanical Gardens); and a bird estuary (Rivulet Terre Rouge Estuary Bird Sanctuary) (Fig. 5) [26]. Spatial information on these protected sites

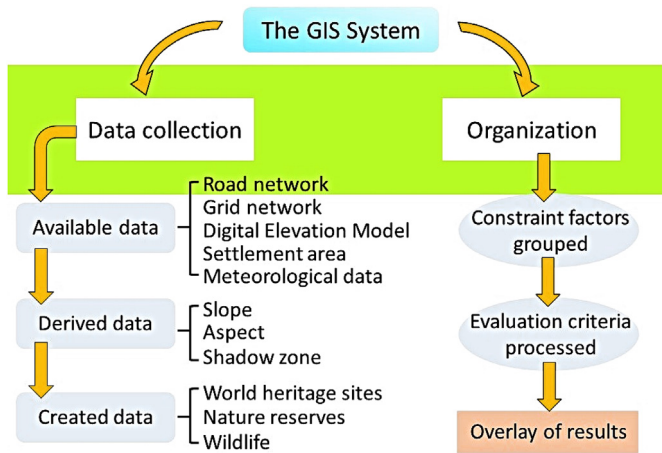


Fig. 3. The GIS system employed to model the solar photovoltaic potential of Mauritius.

have been acquired from the NBSAP and the World Database on Protected Areas (WDPA). The latter is a joint project between the United Nations Environment Programme (UNEP) and the International Union for Conservation of Nature (IUCN), coordinated by UNEP World Conservation Monitoring Centre (UNEP-WCMC).

2.2.4.3. Shadow zones. The shadows cast out by the geomorphological structures of the terrain attenuate significantly the amount of solar radiation received by a region situated in the projected shadow. Near sunrise and sunset, the distance to the end of the shadow is the largest [27]. Consequently, the shadows projected by the variable land slope as the sun rises or is about to set, diminish considerably the exposure of an area situated directly in the shadow zone, to incoming sunlight. The exposure of PV panels to sunrays is of utmost importance since it dictates the amount of electricity output from such a system. A shadow analysis is performed on the 10×10 m DEM of the island made available by the Ministry of Housing and Lands. The elevation and azimuth angle of the sun changes depending on the time of the year. The longest shadows are cast when the sun is at the lowest point in the sky, corresponding to the winter solstice in Mauritius. Subsequently, using ArcMap software, the shadows produced in summer (azimuth angle: 110° and 250°) and winter solstices (azimuth angle: 60° and 300°) corresponding to an elevation of 10° from sunrise and to sunset were derived, with azimuthal angle acquired from the

chart presented in Fig. 4. Additionally, we compute the shadow projected at noon in the winter solstice (elevation: 47° and azimuth: 0°), knowing beforehand that the sun positioned directly overhead of Mauritius at noon in the summer solstice would cast practically no shadow. The raster layers are then superposed with the aim to obtain the outline of the shadow regions throughout the year (Fig. 5).

2.2.4.4. Slope limitations. One technical limitation when it comes to the installation of ground-mounted PV panels is the suitability of the terrain slope. In this study, we analyse the potential of stand-alone PV system. For a stand-alone system, the limitations of the slope on which installations can be made is dependent on the design of the rack. With the objective to explore the PV potential of the island, we will consider the maximum slope that a terrain can accommodate installations on. In a study performed by the University of Arizona to locate optimum sites for PV solar panels, a slope angle of greater than 35° was rejected from the surface analysis [28]. In fact, some companies offer facilities regarding the design of racks for slopes up to 40° [29]. In the present study, we reject any slope greater than 35° for the analysis of PV potential in Mauritius (Fig. 5). Grading is then performed on the slope criteria for classification from best (flat, 0°) to worst (inclined, 35°) pertaining to the placement of ground-mounted PV systems. In general, flat land is preferable for the ease of installation in addition to shadow effects casted on a group of interconnected panels. Land slopes greater than 4% results in shadowing of next row of solar panels which in turn results in diminishing the overall system efficiency [30]. The slope exclusion and grading areas are derived from the DEM of the island.

2.2.5. Evaluation criteria

Criteria regarded as influential and conditional factors for the placement of ground-mounted PV infrastructure are discussed in this section.

2.2.5.1. Electricity generation from PV system. The amount of electricity generated from PV systems is the most important parameter that influences the location within the island where to concentrate such facilities. The conversion of solar energy into electricity goes through a series of phases where losses occur owing to the efficiency of parts making up the PV system. The electricity generated is given by equation (1) [31]:

Table 1

Summary of GIS models employed in literature accompanied with the proposed model as evaluated in various countries.

Model	Technique	Evaluation Criteria	Study Area
Asakereh et al. [17] (2017)	Fuzzy Logic and AHP	Slope; Water body; Agriculture land; Natural reserve; Residential areas; Roads and railroad network; Solar insolation	Iran
Al Garni and Awasthi [18] (2017)	Multi-Criteria Decision-Making and AHP	Solar irradiation; Air temperature; Slope; Land aspect; Proximity to settlement areas; Proximity to road and grid networks	Saudi Arabia
Sindhu et al. [19] (2017)	TOPSIS and AHP	Proximity to settlement areas; Public acceptance; Solar radiation; Infrastructural cost; Wild life and endangered species impact; Land acquisition; Noise impact	India
Majumdar and Pasqualetti [20] (2018)	GIS and Multi-Criteria Analysis	Slope and Aspect of land; Distance from transmission lines; Distance from roads, highways and railways; Solar radiation; Distance from wildlife; Distance from wetlands; Distance from developed areas; Distance from places of cultural and historical importance; Distance from areas for recreational activities	USA
Merrouni et al. [21] (2018)	GIS and AHP	Solar irradiation; Slope; Distance from residential; Distance from road and railway network; Distance from Electricity grid; Distance from water resources	Morocco
Wang et al. [22] (2018)	Multi-Criteria Decision-Making and AHP	Temperature; Sunshine hours; Humidity; Distance from major road; Distance from power network; Protection law; Legal and Regulatory compliance; Potential demand; Ecology; Elevation	Vietnam
Doorga et al. (Proposed model)	Multi-Criteria Decision-Making and AHP	Electricity output derived from solar irradiation and air temperature; Sunshine duration; Slope; Aspect; Proximity to settlement areas; Proximity to main roads; Proximity to grid network	Mauritius

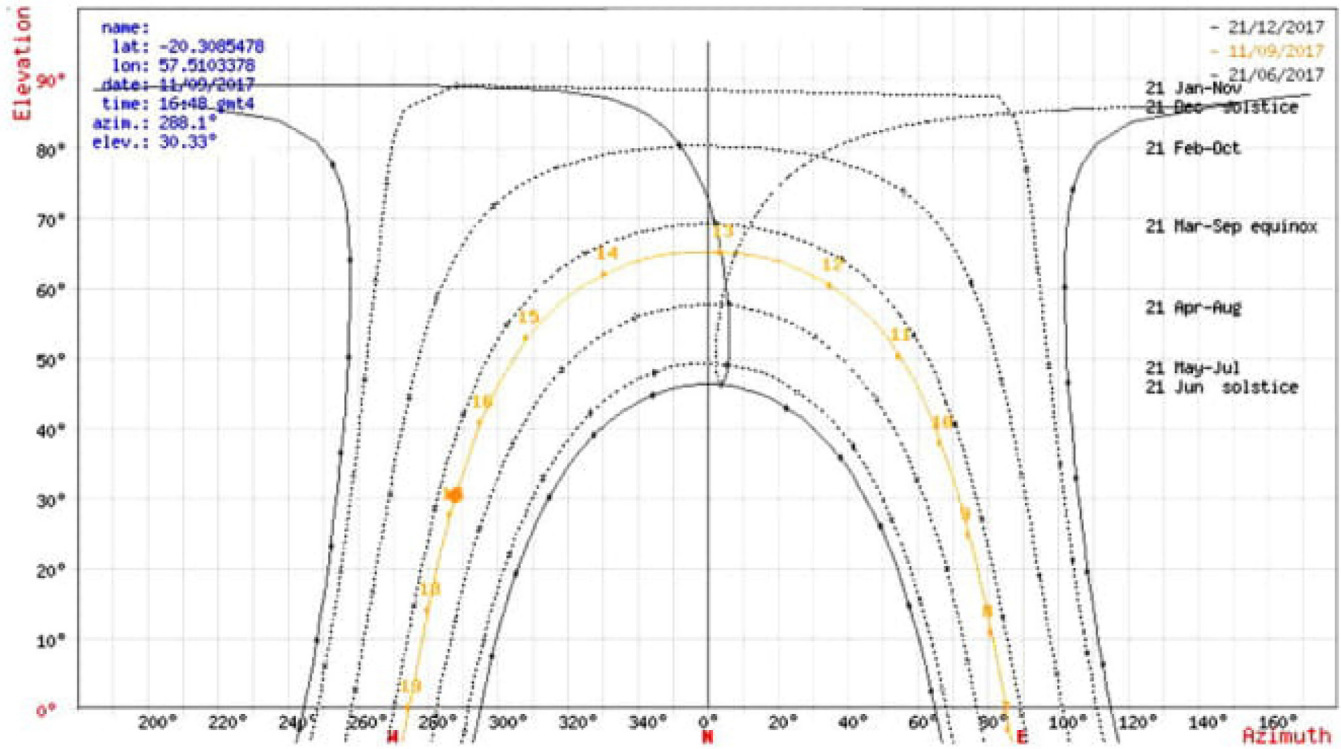


Fig. 4. Elevation against azimuth angle chart for the year 2018 in Mauritius.

$$E_{AC} = G_s \cdot K \cdot \eta_{pv} \quad (1)$$

Where E_{AC} (kWh/m²/month) is the monthly mean daily alternating current generated per unit area by the photovoltaic system; G_s (kWh/m²/month) is the monthly mean daily global solar radiation; K is the performance ratio that includes all loss factors from photovoltaic surface to the output terminal; and η_{pv} is the rated conversion efficiency of commercial wafer-based silicon modules. The last 10 years has seen the efficiency of average commercial wafer-based silicon modules increase from about 12% to 17% [32]. In the study we use a conservative and pragmatic value of 15% for the efficiency of c-Si modules.

The performance ratio of the photovoltaic system is given by equation (2) [33]:

$$K = K_{HD} \cdot K_{TD} \cdot K_{PM} \cdot K_{PA} \cdot K_{PT} \cdot \eta_{inv} \quad (2)$$

Where K_{HD} is the parameter that accounts for losses occurring on the photovoltaic surface; K_{TD} accounts for time-dependent losses by the module function; K_{PM} accounts for losses by the array load; K_{PA} accounts for losses by the array circuit; K_{PT} is the module temperature coefficient; and η_{inv} is the effective energy efficiency of the inverter. The JIS recommended values of 0.97 for K_{HD} , 0.95 for K_{TD} , 0.94 for K_{PM} , 0.97 for K_{PA} , 0.90 for η_{inv} are used [34]. These standard ratios have been used by China to evaluate its photovoltaic resource potential [33]. Furthermore, it has also being employed by the African region of Kenya for assessing its solar photovoltaic potential in health institutions [35]. The ratios have also been employed by Kawajiri et al. (2011) [36] to evaluate the photovoltaic potential of the World.

Solar panels are experimented under Standard Test Condition (STC) of 25 °C with an insolation of 1 kW/m² and temperature values above this threshold will result in drop in output power

owing to the negative temperature coefficient of the semi-conducting material [1]. Therefore, equation (3) takes into consideration the effect of temperature on the electricity generated from the module through the module temperature coefficient [33].

$$K_{PT} = 1 + \alpha_{pmax} \times \frac{(T_{CR} - 25)}{100} \quad (3)$$

Where α_{pmax} is the maximum power temperature coefficient, which is numerically equivalent to -0.41°C^{-1} for c-Si PV modules; and T_{CR} is the temperature of the module which can be derived from equation (4).

$$T_{CR} = T_{AV} + \Delta T \quad (4)$$

Where T_{AV} is the mean monthly daily ambient temperature; ΔT is the weighted average of the annual increase of module temperature. A value of 18.4 °C is used for ΔT corresponding to photovoltaic systems mounted on racks [34].

In an attempt to calibrate the performance of equation (1) for the case of Mauritius, we resort to the typical four-year averaged monthly production of a 15 MW solar power plant located near Medine (20.2° South; 57.4° East). The technical specifications for the panel efficiency and total module area for the solar farm were used. Also, long-term monthly mean meteorological data for the time period 1961–1990 for global solar radiation and ambient temperature near Medine were acquired from the Mauritius Meteorological Services (MMS). A seasonal parameter, ϑ_s , is included in equation (1) for the test case of Mauritius. The seasonal parameter has been obtained by dividing the recorded monthly electricity production of the solar farm by the theoretically derived monthly values from equation (1). Fitting the 5th degree polynomial trend through the set of data points (Fig. 6(a)) for generalizing, resulted in ϑ_s taking values of 1.019 in January; 0.9465 in February;

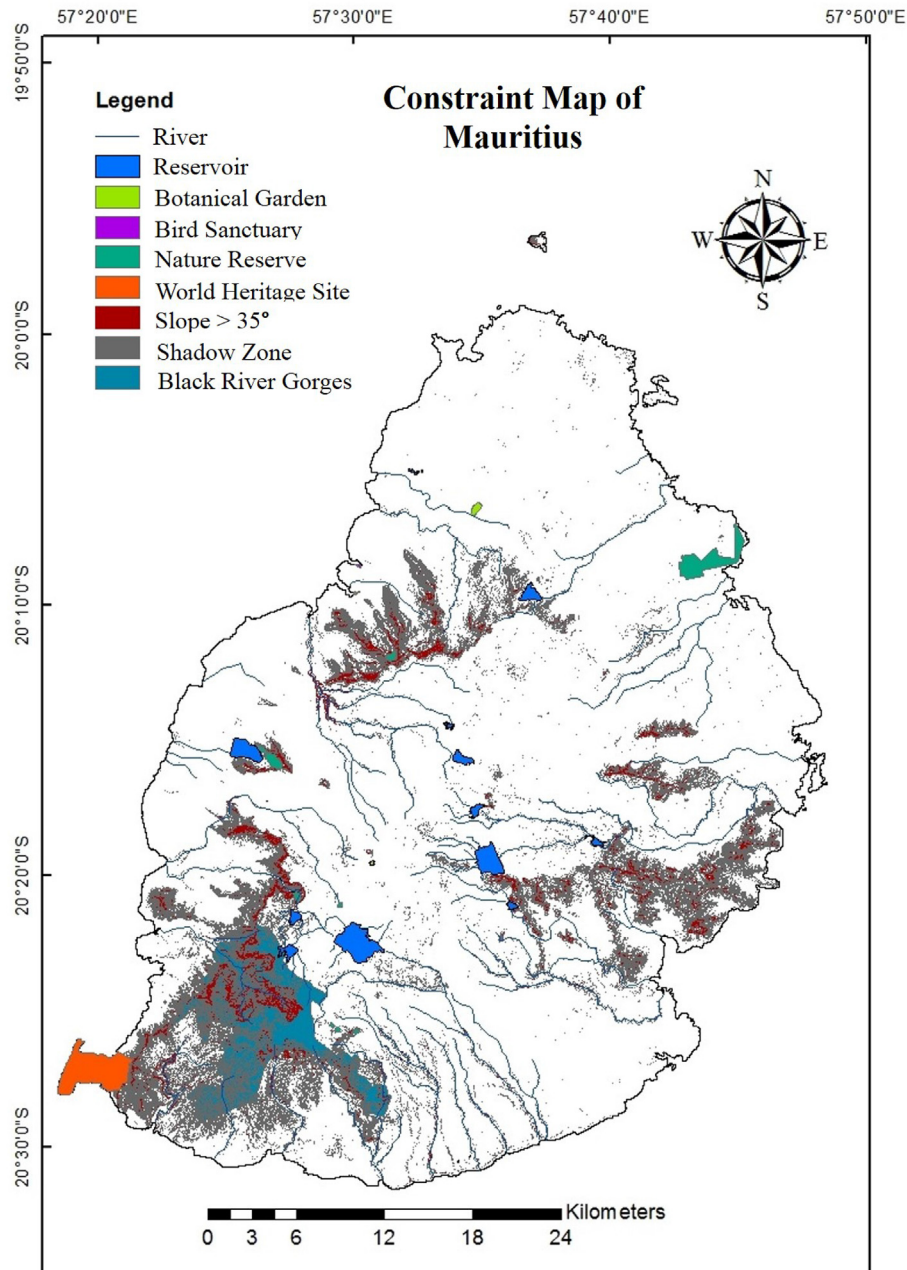


Fig. 5. Map of Mauritius at a scale of 1:300,000 depicting the spatial constraints for PV installation.

1.1150 in March; 1.3280 in April; 1.4810 in May; 1.5338 in June; 1.4860 in July; 1.3730 in August; 1.2480 in September; 1.1210 in October; 1.0390 in November; and 0.9745 in December. From the residual plot of Fig. 6(a), the residuals lie within the $[-0.1, 0.1]$ interval, giving an indication of the relatively good fit. Therefore the proposed amended PV electricity production formula specific to the region of Mauritius is given by equation (5):

$$E_{AC} = G_s \cdot K \cdot \eta_{pv} \cdot \vartheta_s \quad (5)$$

Substituting the seasonal parameter derived in equation (5) resulted in the theoretical derivation of the monthly electricity production for the typical 15 MW solar farm as presented in Fig. 6(b). A strong positive linear correlation (0.8914) is observed between the recorded and theoretical trends. The mean absolute

percentage error between the two trends is observed to be 2.25%, which reflects high estimation accuracy.

The spatial variations in electricity output from PV systems on spatial scales for the island are then derived using equation (5) and coupled to the Inverse Distance Weighting (IDW) interpolation technique to produce the map shown in Fig. 7.

2.2.5.2. Sunshine duration. The duration of sunshine is an important factor when it comes to determining the most suitable region for implementing PV facilities. It determines the exposure of a PV system to sun rays and therefore dictates the quantity of electrical energy output. Sunshine hours vary as a function of latitude and daytime cloudiness [37]. The distinct microclimate systems prevailing at different regions on the island lead to variable heterogeneities associated with cloud system, thereby resulting in

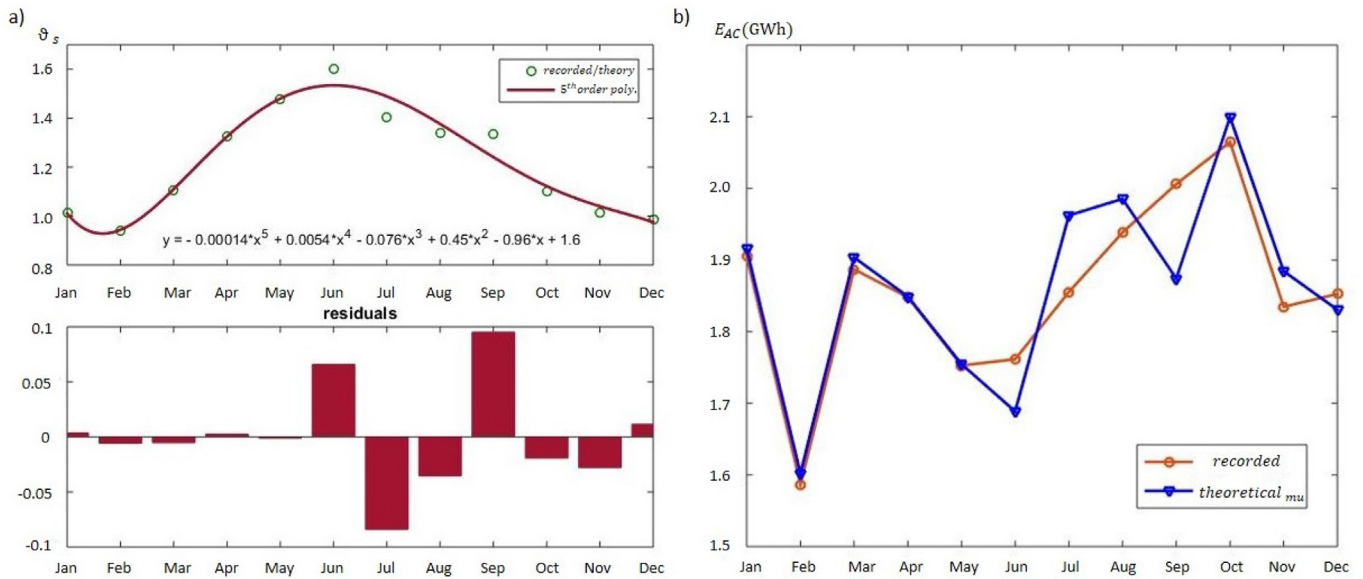


Fig. 6. (a) Seasonal parameter for the scenario of Mauritius and corresponding residual plot. (b) Typical solar farm recorded and theoretical production trends as derived using the seasonal parameter for Mauritius.

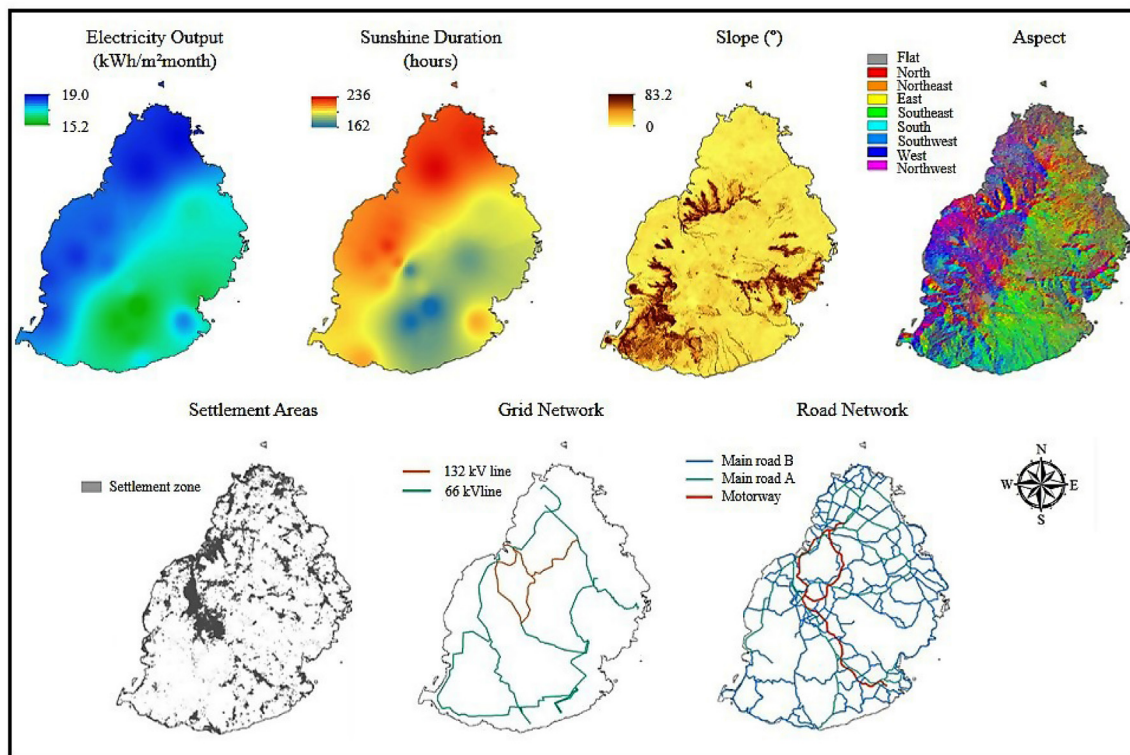


Fig. 7. Spatial information on the evaluation criteria which includes yearly average monthly electricity output and sunshine duration among other criteria such as slope, aspect and proximity factors.

variable sunshine hours. Ground-based measurements of monthly sunshine duration at 17 stations distributed on the island and averaged over the years 1961–1990 have been acquired through the MMS. Interpolation process using the IDW technique has been resorted to produce the sunshine duration map of Fig. 7.

2.2.5.3. Aspect. Aspect estimates the maximum slope direction of

each cell [38]. This factor plays an important role in determining the most suitable parcel of land for the installation of PV panels. One of the main reason contributing to its importance, is the fact that ground-mounted system on which modules are installed, need to be kept fixed. Consequently, it is necessary to orient the system in the direction of maximum exposure to incoming solar radiation. Owing to the geographic position of Mauritius in the southern

hemisphere, land surface facing the geographic north should be prioritized. These surfaces will be more exposed to the sun rays throughout the year than those surfaces facing the other three cardinal points. The aspect map (Fig. 7) is derived from the DEM of Mauritius.

2.2.5.4. Proximity to road. Vehicle access to and from the construction site offers great importance for building and maintenance purposes. The proximity of the most suitable site for PV placement from existing road network has associated economic and environmental benefits. The economic factor includes avoiding additional costs for road construction while the environmental factor aims at the reduction of environmental damage that the construction of such a project could entail [39]. The digitized road network (Fig. 7) has been available from the Mauritius Road Development Authority.

2.2.5.5. Proximity to grid lines. The focus of the present paper is on evaluating the PV potential of stand-alone ground-mounted systems. Nonetheless, if the case is to be broadened to large-scale interconnected power requirements, access to transmission grid is of paramount importance due to the high voltage demand being dealt with [39]. The construction of new power transmission systems generally entail high costs. Consequently, using the available power transmission system would not only cut down on capital costs of the project, but also reduce power losses arising from the transmission of electricity through extended distances. The digitized high voltage transmission lines (Fig. 7) of the country has been made available from the Central Electricity Board.

2.2.5.6. Proximity to settlement areas. It is more favourable to generate electricity near big cities inhabited by a large population. Short distance between supply and demand not only reduces the congestion in the transmission network but also reduces power losses and minimises transmission cost by avoiding the need for lengthy and costly transmission lines [39]. Coupled with the proper storage unit, solar energy systems can additionally serve as a backup electricity supply to large cities in case of power failure from conventional network. The settlement areas of the country (Fig. 7) has been made available from the Ministry of Housing and Lands.

2.2.6. Theoretical background

2.2.6.1. Analytical Hierarchy Process. The Analytical Hierarchy Process (AHP), developed by Saaty in the 1970s, is described as a framework of logic and problem resolving that is achievable through the organisation of perceptions, feelings, judgements and memories into a hierarchy of forces that influences decision results [40]. This technique enables complex problems to be solved in a structured manner by breaking them into smaller constituent subparts based on reasoning sub-goals. Instead of making a 'correct' decision, AHP technique guides the decision maker to seek the one that best suits his perception of the problem [41].

The application of this method is inter-disciplinary and touches wide-ranging of fields. Bunruamkaew and Murayam [42] used AHP technique to produce a site suitability analysis of potential ecotourism sites in Surat Thani Province, Thailand. Otay et al. [43] used AHP technique to assess healthcare institutions. Riahi and Moharrampour [44] employed this method to evaluate strategic management in business. Bitarafan et al. [45] used AHP technique to perform architectural evaluation of building's indices in explosion crisis management. Da Rocha et al. [46] resorted to the AHP technique to analyze the operational performance of Brazilian airport terminals. The main steps of this technique are highlighted below:

Step 1: Defining the problem to be addressed.

Step 2: Structuring the problem into a hierarchy of levels based on related criteria.

Step 3: Constructing a set of pairwise comparison matrices for the individual criteria.

Step 4: Computing the maximum eigenvalue, consistency index, consistency ratio of each comparison matrix and determine priority weights associated to each criteria.

A comparison scale (Table 2) ranging from 1 to 9 is used in the AHP, with a value of 1 indicative of two criteria having equal importance and a value of 9 reflecting the strongest order of difference between two criteria.

As an illustration of the theoretical background of the AHP technique, we consider the set of criteria $Z = Z_j (j = 1, 2, \dots, n)$. The result of the pairwise comparison on the set of n criteria can be represented in an $n \times n$ evaluation matrix A . Every element in the matrix A , $a_{ij} (i, j = 1, 2, \dots, n)$ is the quotient of the weights of the criteria as presented in equation (1):

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \quad (6)$$

Where $a_{ii} = 1, a_{ji} = \frac{1}{a_{ij}}, a_{ij} \neq 0$. The relative weights corresponding to each matrix, determined after the normalization process, are given by the right eigenvector (v) corresponding to the largest eigenvalue (λ_{max}), as follows:

$$Av = \lambda_{max}v \quad (7)$$

If the pairwise comparisons are absolutely consistent, the matrix A has rank 1 and $\lambda_{max}v = n$. When such a condition is met, the weights can be deduced by normalizing any of the rows or columns of A .

The quality of the output of the AHP is strictly related to the consistency of the pairwise comparison judgements. The relation between the entries of A ($a_{ij} \times a_{jk} = a_{ik}$) and the consistency index (CI) is given by:

$$CI = \frac{\lambda_{max} - n}{n - 1} \quad (8)$$

The conclusion pertaining to whether the evaluations are sufficiently consistent, as determined by the final consistency ratio (CR), is deduced through the ratio of the CI to the random index (RI) as shown below:

$$CR = \frac{CI}{RI} \quad (9)$$

In order to be acceptable, the CR index must be lower than 0.10. However, if the final consistency ratio exceeds this value, the procedures need to be repeated with the aim to improve the consistency.

The same scale has been adopted by Doljak and Stanojević (2017) in finding the natural conditions pertaining to locating the optimum site for ground-mounted photovoltaic power plant in Serbia [47]. Similarly, Aly et al. (2017) employed the same scale for identifying photovoltaic hotspots in Tanzania [39]. Finally, the same weighting scale has been used by Uyan (2013) for solar farm site selection in Turkey [48].

2.2.6.2. Weighted Linear Combination. Weighted Linear Combination (WLC) is based on the concept of weighted average in which continuous evaluation criteria identified are standardized to a

Table 2

Nine-point intensity of importance scale with accompanying definition and implication.

Weight	Definition	Implication
1	Equal importance	Two criteria contribute equally to objectives
3	Moderately more important	One criteria slightly favoured over another
5	Strongly more important	One criteria strongly favoured over another
7	Very strongly more important	One criteria favoured very strongly over another
9	Extremely more important	The evidence favouring one criteria over another is of highest possible order of affirmation
2, 4, 6, 8	Intermediate values	Employed to represent compromise between the priorities listed above
Reciprocals of above intensity of weights		If criteria i has one of the above non-zero numbers assigned to it when compared to criteria j, then j has the reciprocal value when compared with criteria i

common scale range prior to performing a combinatorial process using the weighted average technique [49]. This technique has been extensively employed in inter-disciplinary areas and have been proved as being a reliable tool. Shahid [50] used this method to assess the impact of climate change on irrigation water demand of dry season Boro rice in north-west Bangladesh. Also, Ayalew et al. [51] employed this method for landslide susceptibility mapping in Japan while Al-Hanbali et al. [52] used this technique to select optimum solid waste disposal sites in Jordan.

The weight assigned to each evaluation criteria layer and acquired through the AHP process is multiplied by the scaled value of the functional parameter before performing a summation of the products over all parameters identified. The scores resulting from such a mathematical operation on each cell of the two dimensional top view surface of the island are used to produce the map delineating the most suitable regions for development. Equation (10) describing the WLC process is as follows:

$$S_i = \sum_{j=1}^n w_j \cdot x_{ij} \quad (10)$$

Where S_i is the net suitability index of cell i , w_j represents the relative importance weight of criterion j , x_{ij} is the value of cell i under criterion j and n indicates the maximum number of criteria.

2.2.6.3. Inverse Distance Weighting. The Inverse Distance Weighting (IDW) approach is a local deterministic method that computes the value as a distance-weighted average of sampled points in a defined neighborhood [53]. The principle behind the IDW technique is that it takes into consideration the closeness of points to a particular location by assigning a higher degree of influence to points lying closer and then weighing the sample points with inverse of their distance from that location. This method can be used by a state in which there are at least 14 sample points with an appropriate dispersion of data points in local scale levels [54]. IDW has the advantage of being a relatively simple method and therefore can be implemented and executed easily without reducing the quality of the final results [55]. IDW is computed as follows:

$$H_e = \frac{\sum_{i=1}^n H_{s,i} d_i^{-p}}{\sum_{i=1}^n d_i^{-p}} \quad (11)$$

where H_e is the estimated variable parameter at a given point i , $H_{s,i}$ is the sample value at point i , d_i is the distance of the estimated point to the sampled point, n is the number of sampled points used for the estimation and p ($p = 2$) is the power parameter that defines the rate of reduction of weight with distance.

2.2.7. Map pre-processing and reclassification

Based on the number of evaluation criteria selected which are thought to influence the placement of the optimal location for PV

harnessing facilities, a number of pairwise comparisons should to be performed. In the present study, a list of 7 evaluation criteria has been targeted, resulting in 21 pairwise comparisons. The criteria comprise of:

- Electricity from PV system (Z_1)
- Sunshine duration (Z_2)
- Slope (Z_3)
- Aspect (Z_4)
- Proximity to settlement areas (Z_5)
- Proximity to grid (Z_6)
- Proximity to road (Z_7)

Based on the 21 comparisons performed, a consistency ratio of 0.058 has been obtained. Since this value is less than 0.10, the consistency test was successful. The decision matrix is shown in Table 3. The resulting weights depend on the principal eigenvector of the decision matrix corresponding to a principal eigenvalue numerically equivalent to 7.467 while the eigenvector solution comprises of 6 iterations, giving a delta value of 9.7×10^{-9} . The resulting weights attributed to individual evaluation criteria together with the associated grade of reclassified raster layers are presented in Table 4.

The weights resulting from the AHP process for each criteria are as follows: Electricity from PV (43.3%); sunshine duration (19.3%); slope (16.6%); aspect (8.9%); proximity to settlement areas (5.2%); proximity to grid (3.9%); proximity to road (2.8%). The weights attributed to corresponding factors will be used in the overlaying process.

Prior to performing the weighed overlay technique, an important step involves the reclassification of the evaluation criteria. The reclassification process converts the different criteria layers into integer raster maps having a common scale from one to nine. We resort to the 'natural breaks' classification method to do the reclassification. This method classifies data values into natural groupings statistically and is designed in such a way as to determine the best arrangement of values into different classes [28].

The WLC technique is then performed through the overlaying of reclassified criteria as illustrated in Fig. 8. The process results in a raster layer containing both suitable and unsuitable sites. The spatial extents regarded as constraints to the exploitation of PV

Table 3

Decision matrix based on a combination of ten pairwise comparisons.

	Z_1	Z_2	Z_3	Z_4	Z_5	Z_6	Z_7
Z_1	1	3.00	5.00	6.00	7.00	7.00	8.00
Z_2	0.33	1	2.00	3.00	4.00	4.00	5.00
Z_3	0.20	0.50	1	3.00	5.00	5.00	6.00
Z_4	0.17	0.33	0.33	1	3.00	3.00	4.00
Z_5	0.14	0.25	0.20	0.33	1	2.00	3.00
Z_6	0.14	0.25	0.20	0.33	0.50	1	2.00
Z_7	0.12	0.20	0.17	0.25	0.33	0.50	1

Table 4

Criteria and factor weights with the corresponding grades as used in the model.

Criteria Weight	Criteria	Factor Weight	Factor	Classes	Grade
0.626	Climate-related	0.433	PV output ($kWh/m^2 month$)	18.56–18.96	9
				18.25–18.56	8
				17.96–18.25	7
				17.64–17.96	6
				17.31–17.64	5
				16.96–17.31	4
				16.60–16.96	3
				16.15–16.60	2
				15.23–16.15	1
		0.193	Sunshine duration (hours)	226.28–235.99	9
				220.00–226.28	8
				213.71–220.00	7
				207.88–213.71	6
				201.65–207.88	5
0.255	Topography	0.166	Slope (%)	195.05–201.65	4
				188.11–195.05	3
				178.98–188.11	2
				162.19–178.98	1
				0–1.31	9
				1.31–4.24	8
				4.24–7.83	7
				7.83–12.40	6
				12.40–16.65	5
				16.65–20.57	4
		0.089	Aspect	20.57–24.16	3
				24.16–28.75	2
				28.75–35.00	1
				North	9
				Northeast	7
				East	5
				Southeast	3
				South	1
				Southwest	3
				West	5
				Northwest	7
0.119	Location	0.052	Proximity to settlement areas (m)	800–900	9
				700–800	8
				600–700	7
				500–600	6
				400–500	5
				300–400	4
				200–300	3
				100–200	2
				0–100	1
		0.039	Proximity to grid (m)	0–998.0	9
				998.0–1996.4	8
				1996.4–2994.7	7
				2994.7–3993.0	6
				3993.0–4991.2	5
				4991.2–5989.4	4
				5989.4–6987.6	3
				6987.6–7985.8	2
				7985.6–9982.3	1
		0.028	Proximity to road (m)	0–433.3	9
				433.3–866.7	8
				866.7–1300.0	7
				1300.0–1733.3	6
				1733.3–2166.7	5
				2166.7–2600.0	4
				2600.0–3033.4	3
				3033.4–3466.7	2
				3466.7–4500.0	1

potential is then removed from the map. The resulting surface contains regions within the island that are regarded as high potential sites for PV infrastructure deployment.

3. Results

3.1. Solar PV potential

The solar PV potential map of Mauritius is illustrated in Fig. 9.

The map depicts the yearly mean monthly solar PV potential which has been acquired through the superposition of yearly mean monthly climate-related factors and taking into account topographical and proximity criteria. A grading system from excellent (grade A) to mild (grade I) have been used to classify the various regions on the island deemed exploitable for PV infrastructure deployment. The map reveals the higher solar PV potential of the northern region districts of Pamplemousses and Riviere du Rempart, both having sites predominately attributed grades A and B.

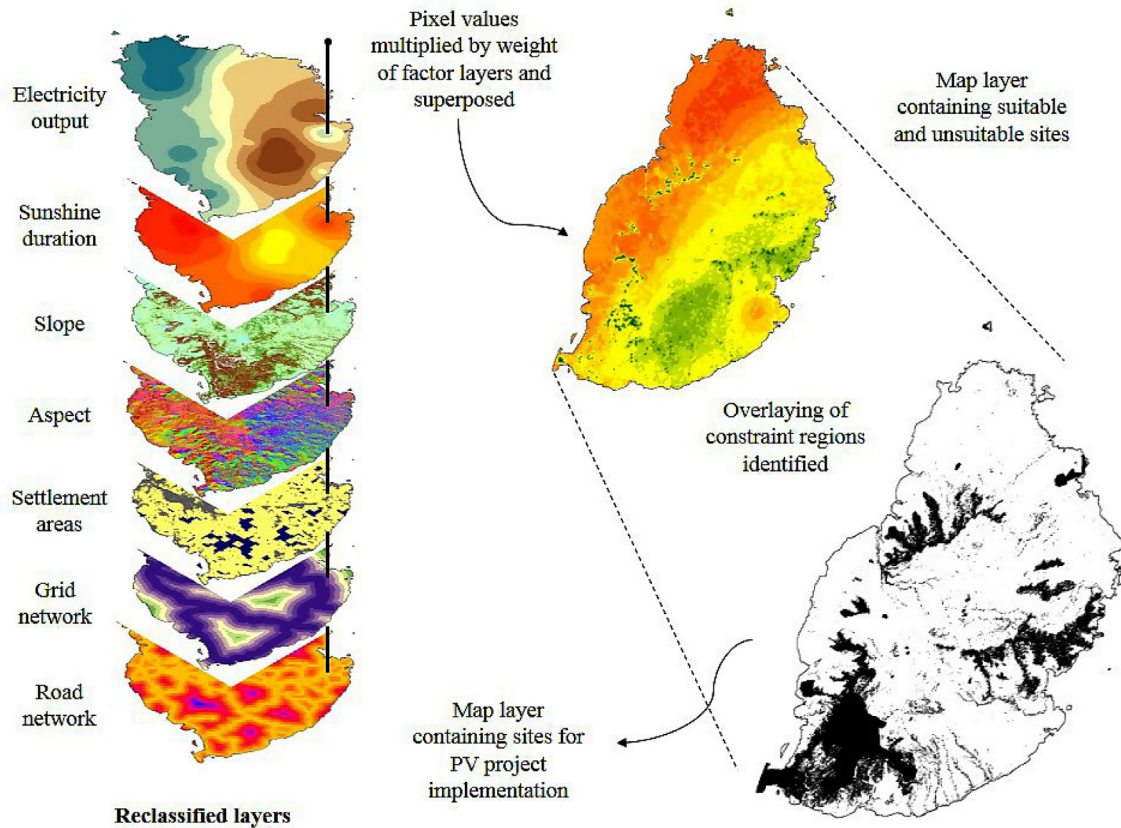


Fig. 8. Illustration of the reclassification and overlaying processes culminating into a map layer containing sites for PV project implementation.

The favorable solar PV potential of these regions is mainly due to the higher electricity output and sunshine duration prevailing. These two criteria have been observed to contribute to a higher weightage (criteria weight of 0.626) in the final raster layer than the other influential criteria. Consequently, the promising microclimate systems prevailing in these regions give rise to the suitable level of temperature, solar radiation and sunshine duration for the construction of PV facilities.

The western region (district of Black River) of the island as well as the northwestern part of the central plateau (Plaines Wilhems and Moka) are predominately grade D type with a significant amount of grade C. Additionally, the southeastern region near Plaine Magnien is classified as grades C to D. The upper southern district of Savanne and lower part of the central plateau have an intermediate to mild solar PV potential. This is mainly attributed to the extended highest rainfall isohyet observed over these regions [56]. Numerous instances exist which highlight the association between orography and precipitation patterns on spatial scales. It has been reported that for large mountain ranges, precipitation maximizes over the windward slopes, whereas for smaller hills the maximum tends to occur nearer the crest [57]. As illustrated in Fig. 1(a), the landform increases in elevation from the southeast to the central plateau. The persistent trade winds from the southeast causes forced orographic uplift of incoming oceanic air over the southeastern slope of the island which results in precipitation near the southern and eastern edges of the central plateau.

Much of the spatial variability in the photovoltaic suitability of Mauritius is attributed to the spatial distribution of solar irradiation. The spatial variations in the solar regime of the island on microscale levels were described by Doorga et al. (2018b). It was

observed that the lowest global solar radiation intensities were recorded in the southern plains of the central plateau which was attributed to the clustering of cloud cover zones formed by forced advection due to the topographic relief. Unsaturated but humid oceanic air carried by the South East Trade Wind is forced to rise due to the rising topography and in the process cools at the dry adiabatic lapse rate. The rising and cooling air reaches its dew point and subsequently becomes saturated at the lifting condensation level whereby the saturated air condenses into a multitude of liquid droplets, with the release of latent heat and causing the air temperature to decrease at the moist adiabatic lapse rate [58]. Due to the release of latent heat, the rising air becomes warmer and unstable as compared to its environment and results in the formation of larger cumuliform clouds. As the southern to southeastern central plateau regions have a steep increasing gradient leading to the highest elevations of the island, this explains much of the cloud cover observed in that specific zone. This in turn causes low insolation and sunshine duration, classifying it as mild and unsuitable for the installation of PV facilities.

As the air at the top of the elevated region of the central uplands continues its journey and is forced to descend on the leeward side of the island, the sinking air increases at the dry adiabatic lapse rate, leaving the cloud formations on the windward side. The descending air is warmer due to latent heat being converted into sensible heat during the condensation process occurring on the windward side [58]. Consequently, the northern and northwestern plains of the island witness relatively high levels of insolation. Additionally, it can also be observed from Fig. 9 that the majority of the high solar PV potential sites of the northern plains are located in the uninhabited zones of the island (Fig. 1(c)). Consequently, this leaves

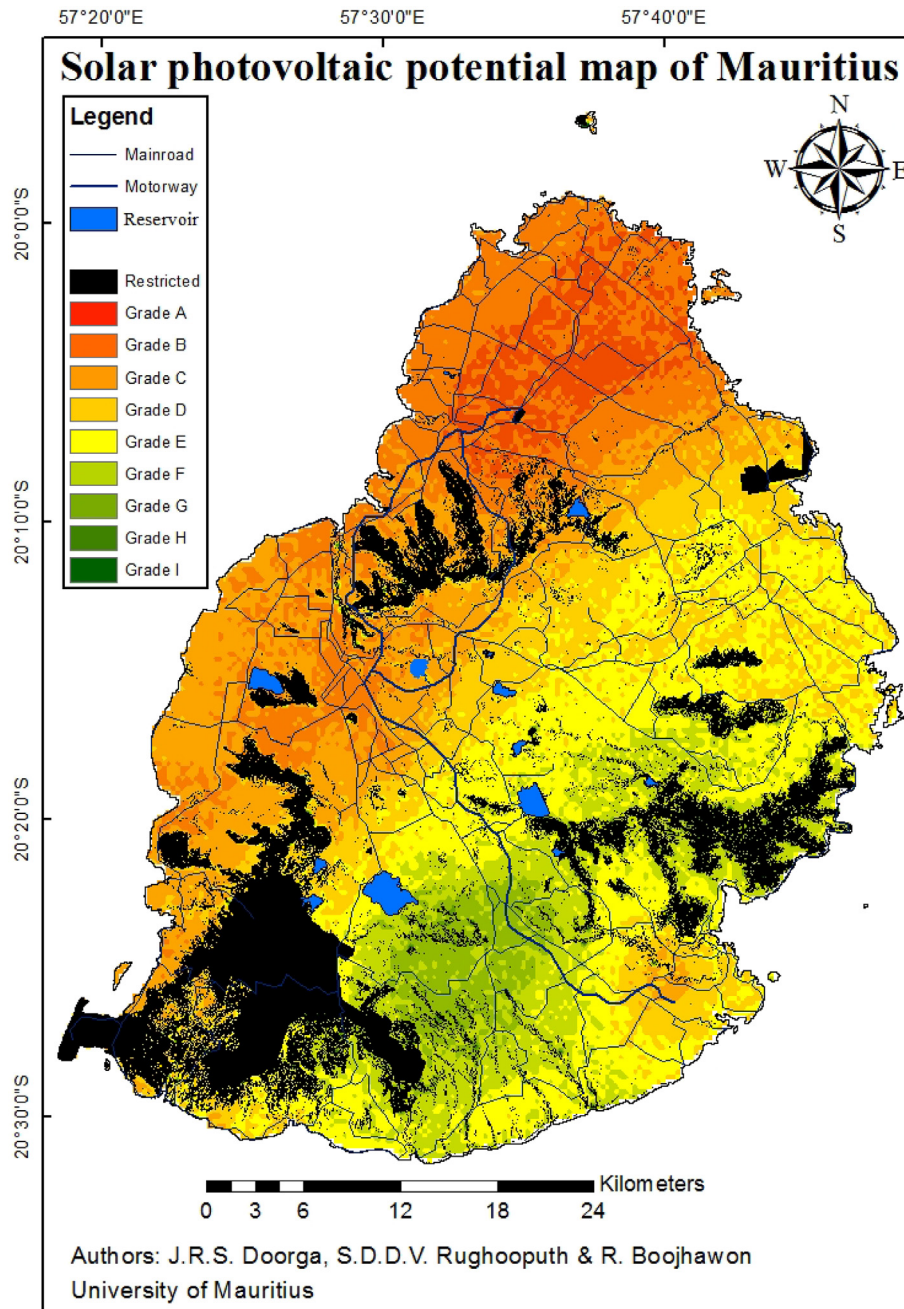


Fig. 9. Solar photovoltaic potential map of Mauritius at a scale 1:300,000.

room for the promoters of the solar PV projects to come forward with an appropriate spatial planning strategy in order to construct efficient PV facilities that will be localized and can be implemented with minimum obstruction or displacement of local residents.

The temporal variations of solar PV potential is explored as depicted in Fig. 10. The monthly maps have been obtained through the superposition of monthly mean daily climate-related factors and taking into account the same topological and proximity criteria as for the yearly map. During the summer months of January to March, the Inter Tropical Convergence Zone (ITCZ) is closest to Mauritius and warm season depressions influence the rainfall climate on the island [59]. It is discernible from Fig. 10 that the greatest contrast (showing the whole spectrum from grade A to grade I) between favorable and less favorable sites, combined with

the largest spatial coverage of mild conditions near the lower central plateau, occurs for the month of January. This effect is attributed to the higher precipitation registered for the month. Mean rainfall in early rainy season summer months of November and December is the lowest throughout the island [60]. As delineated in Fig. 10, the months of November and December witness the lowest contrast (showing only part of the spectrum from grade B to grade G). The lowest spatial variations in solar PV potential occurs in the winter season from May to October, depicting rather stable climatic conditions prevailing on spatial scales. The most promising month for solar photovoltaic potential on spatial scales is observed to be October, with a prevalence of intermediate (F) to very good (B) grades. However, irrespective of the month of the year, modelling of the photovoltaic resource potential of Mauritius

Monthly Solar Photovoltaic Potential Maps of Mauritius

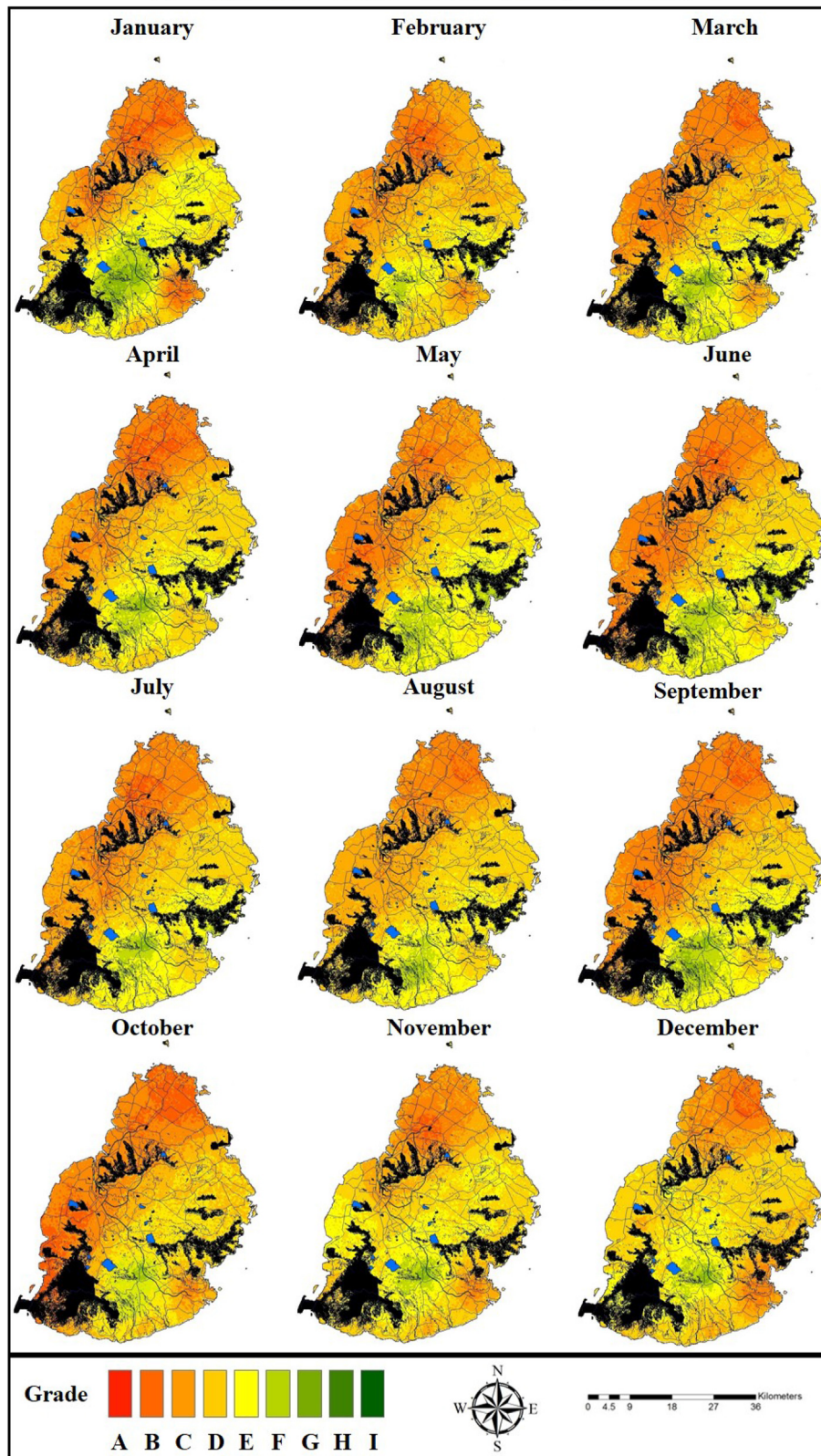


Fig. 10. Spatio-temporal variability of monthly mean yearly solar photovoltaic potential in Mauritius.

gives an indication of the propitious nature of the northern plains.

The identification of the most appropriate region on the island for photovoltaic construction has been identified whilst a temporal analysis of the best region throughout the year has resulted in insignificant variations on spatial scales. Therefore, the results are expected to save on land resources through the construction of solar panels on limited land area thereby encouraging the efficient harness of solar energy. The objective of the study has therefore been accomplished.

With recent interests from the Government of Mauritius in promoting solar photovoltaic projects in an attempt to reduce dependence on remote international markets, this research comes in line with the contemporary vision of self-sufficiency in the energetic needs of a small island developing state. This paper is the first of its kind in Mauritius and would provide valuable information to the research community as well as potential investors pertaining to the most suitable location for installing solar photovoltaic systems. The temporal aspect of solar variability on spatial scales has also been explored so that proper maintenance strategies may be devised for potential solar farms that would be constructed on the island.

3.2. Comparison and validation of results

A high degree of similarity exists between the solar photovoltaic potential map generated in this study through modelling, and the one made available by Solargis under a contract to the World Bank (<http://globalsolaratlas.info/>). In both maps, the northern plains of the island witness high resource potential whilst the lower central plateau region is observed to record the lowest solar photovoltaic resource potential. Another model developed by PVGIS under the European Commission Joint Research Centre (<http://re.jrc.ec.europa.eu/>) shows the same resemblance with the model developed in this paper. However, the model developed using the presented methodological framework has the advantage of yielding more accurate results. The model implemented is therefore validated.

3.3. Distributed PV systems

The Central Electricity Board (CEB) of Mauritius has the task of preparing and conducting development schemes with the objective of promoting, coordinating and improving the generation, transmission and distribution of electricity on the island. The transmission network of the country consists of sixteen major substations and 300 km of single-circuit transmission lines [61].

Integration of solar energy into the utility grid of the island may be at the transmission or distribution levels depending on the capacity of the photovoltaic system. Large-scale solar energy generation is injected to the transmission system whilst medium and small-scale photovoltaic systems are respectively branched to the medium and low voltage distributed systems. However, the intermittent nature of solar photovoltaic systems causes time-varying electricity generation which increases the frequency deviation that may result in load shedding and risk of total system breakdown. As such, the solution to deal with such variable energy sources is to either use a reliable battery storage system or increase the amount of spinning reserve in order to maintain the system frequency within acceptable limits.

Due to a lot of distributed photovoltaic systems, access to high voltage transmission lines is based on the operating voltage requirements. Access to the electricity distribution lines is well established by the regulating authorities of the country. The substations are relevant for utility scale photovoltaic systems of capacity 10 MW onwards. Photovoltaic systems of capacity 4 MW up

to 10 MW can be injected to the 22 kV transmission line into the CEB's 22 kV busbar [61]. Photovoltaic systems of capacity 50 kW to 4 MW can be connected to medium-voltage feeders under the Medium-Scale Distributed Generation (MSDG) scheme. Systems lower than 50 kW can be tapped by low-voltage feeders under the Small-Scale Distributed Generation (SSDG) scheme.

4. Conclusions

This paper analysed the spatio-temporal variability of the solar photovoltaic potential of Mauritius. A Multiple Criteria Decision Making (MCDM) analysis coupled with an Analytical Hierarchy Process (AHP) have been performed in the GIS environment in order to map the solar photovoltaic potential of Mauritius from a list of social, technical, legal, environmental and climatological factors comprising of constraint, conditional and influential factors.

The results showed that the northern region of the island is suitable for the placement of photovoltaic systems. Comparison of the results with that of Solargis and PVGIS reveal that high degree of similarity exists between the maps, thereby validating the model investigated. A temporal analysis conducted throughout the year showed that the suitability of the northern region remained practically unchanged on spatial scales. Additionally, access to high voltage transmission lines is presented for the case of distributed photovoltaic systems.

The results presented in this paper may serve as a strong recommendation to the Government of the island and other relevant authorities to focus photovoltaic system deployment in the northern region of the country so that solar energy may be harnessed efficiently whilst saving land space. The approach adopted in this research is an innovative one and may serve as a methodological framework to other countries wishing to map their solar photovoltaic potential.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.energy.2018.12.072>.

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